Introduction to Computing (CS 101)

Introduction to Von Neumann Architecture

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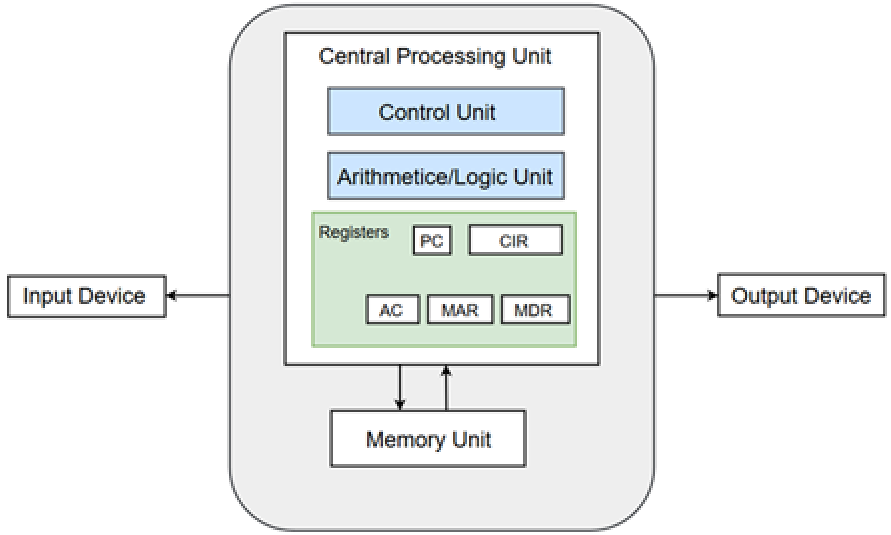
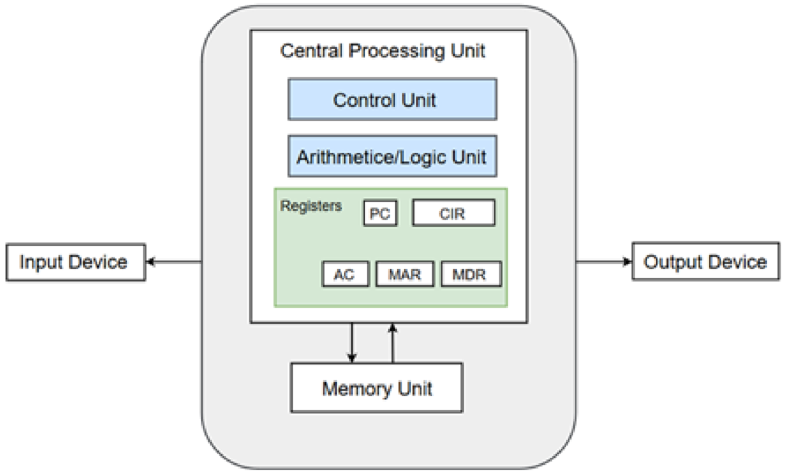
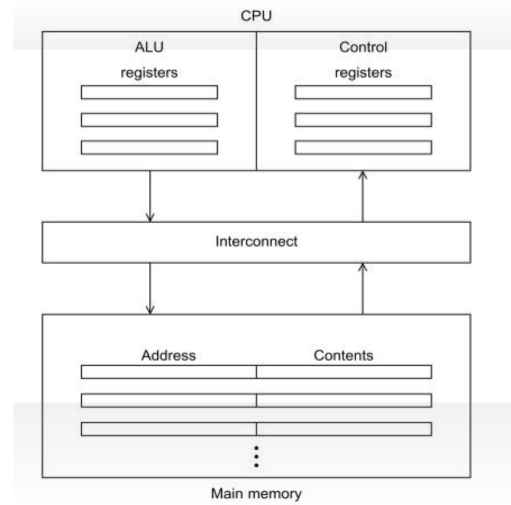
Von Neumann Architecture



- The program instructions are stored in memory and are fetched upon requirement to the CPU for executing.
- Memory is a collection of locations for storing instructions and data.
- Each location has a unique address which is used to access the location.

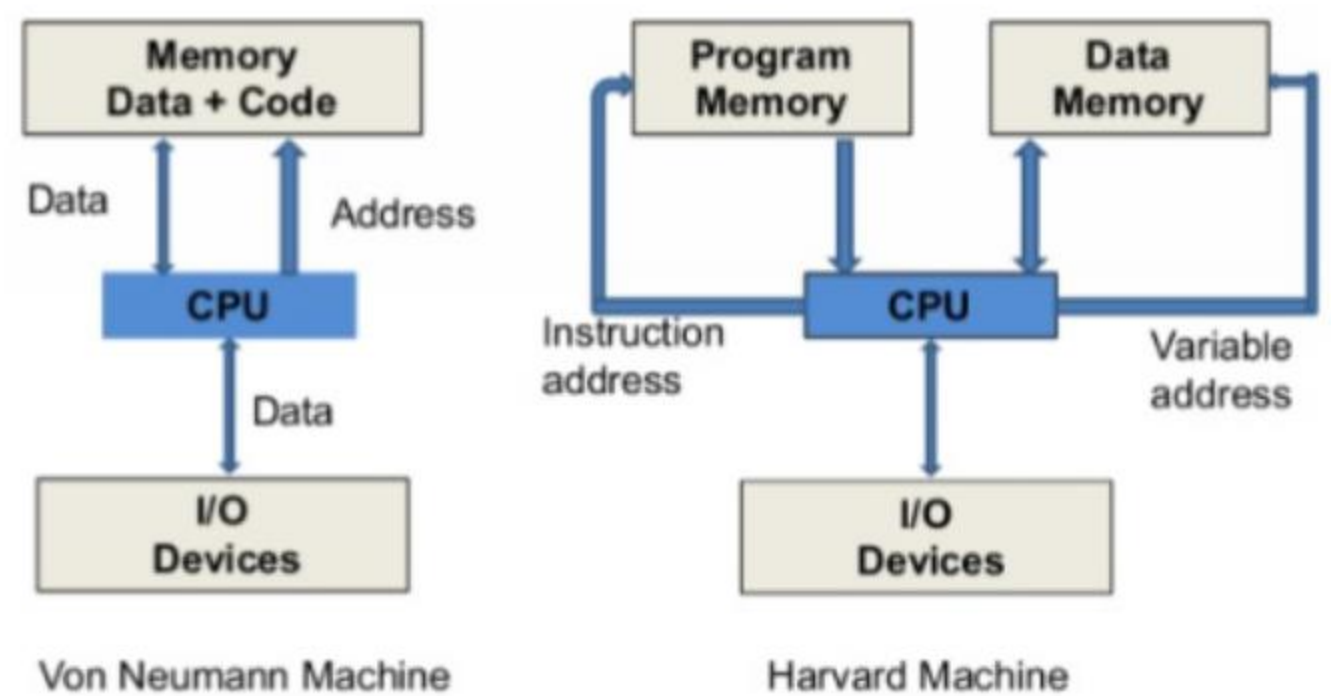
CPU Registers



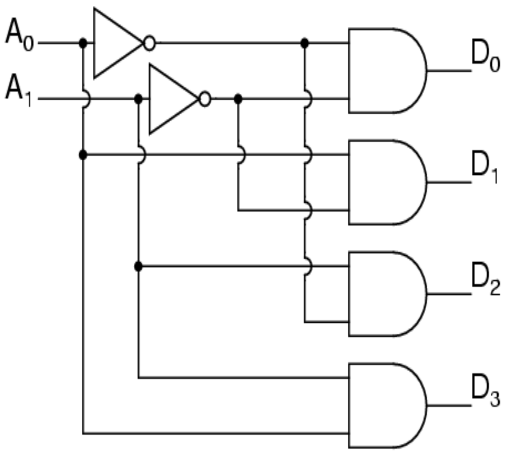


- Registers are very fast storage, residing inside the CPU.
- Registers can be general purpose (integer registers and floating point registers) as well as special purpose (PC, CIR, AC, MAR, MDR etc.)
- General purpose registers store the most frequently used data.
- Data/instruction is transferred between main memory and CPU through an interconnect mechanism (bus/wires).

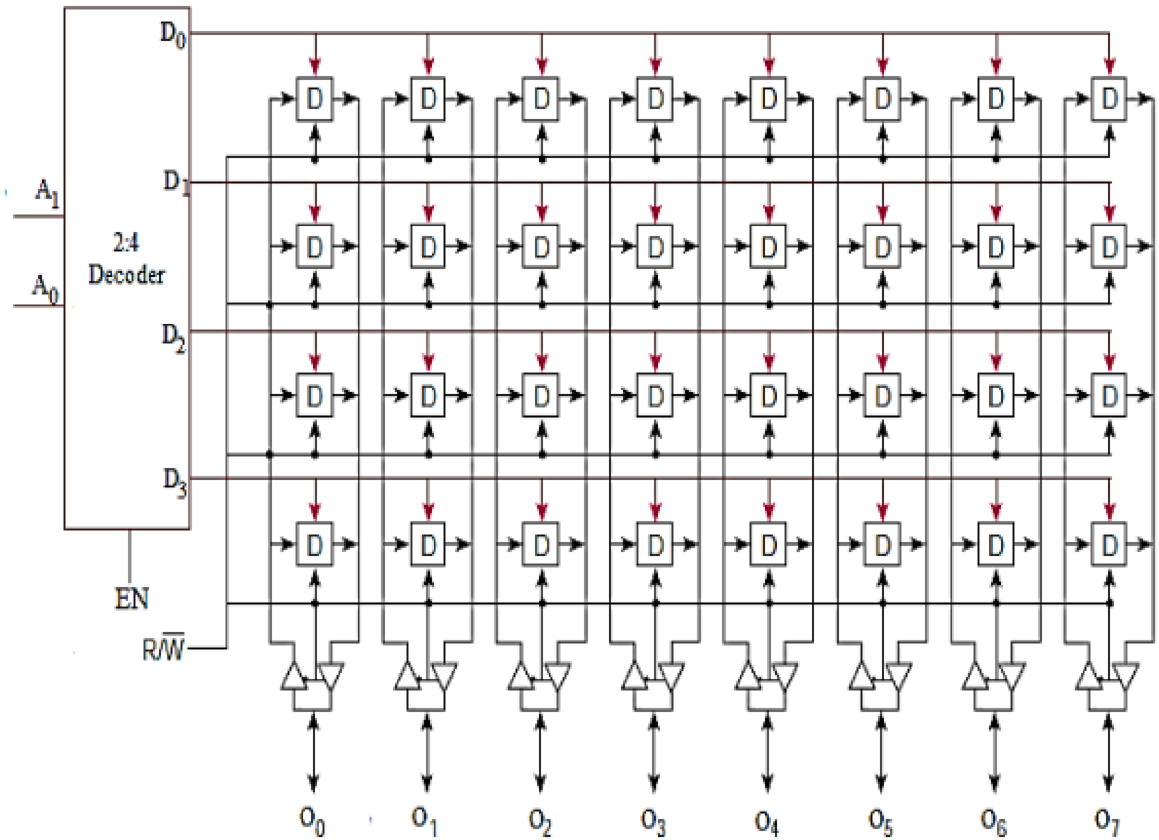
Von Neumann Vs Harvard Architecture



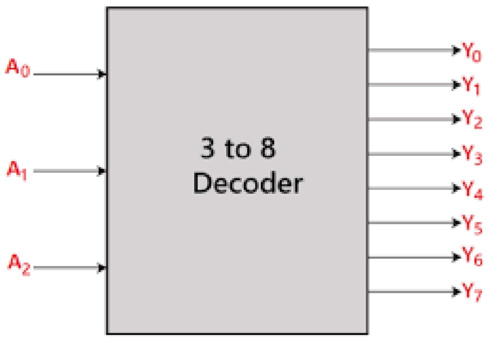
How memory works?

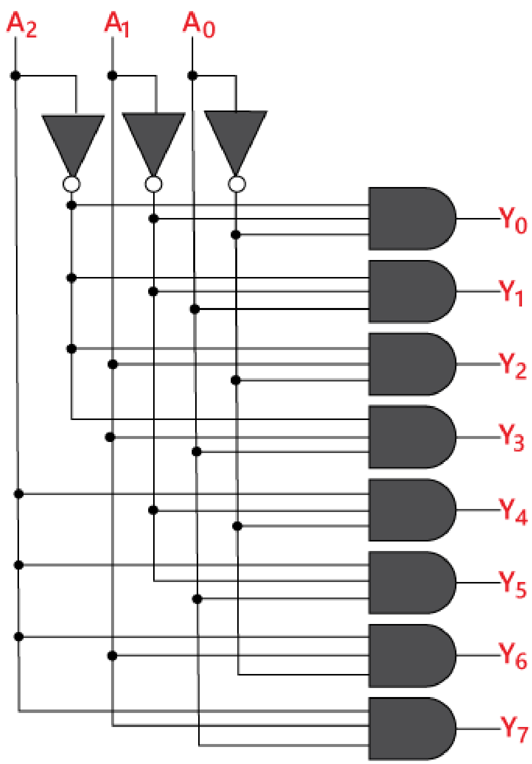


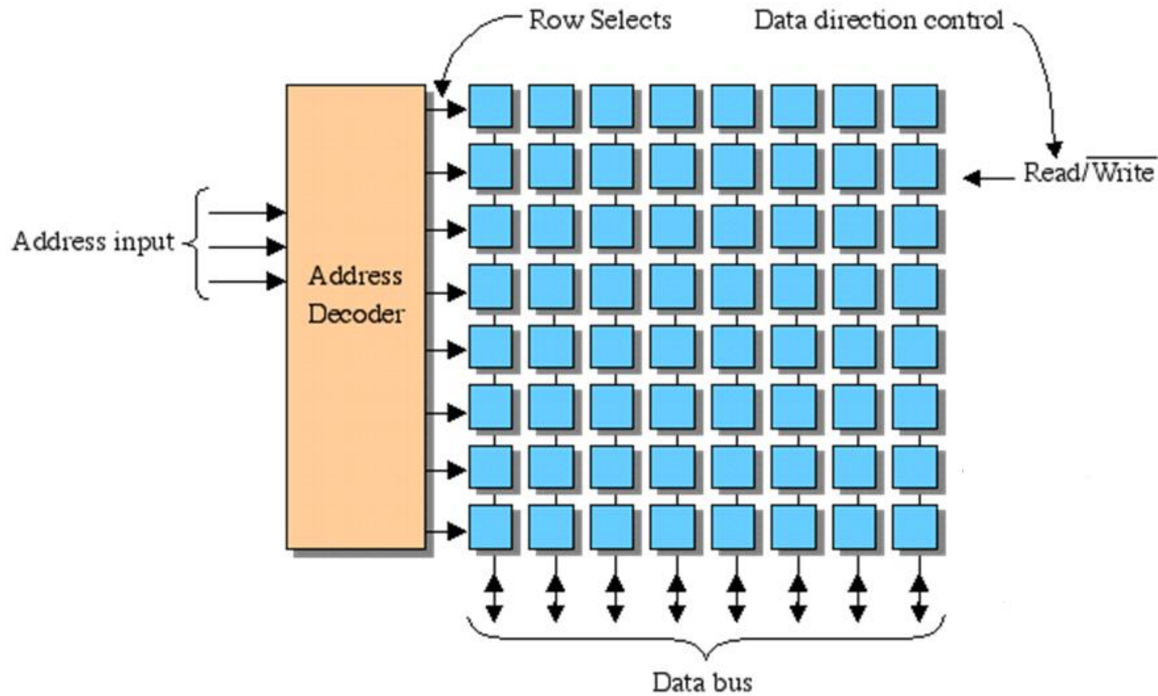
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Address Decoders

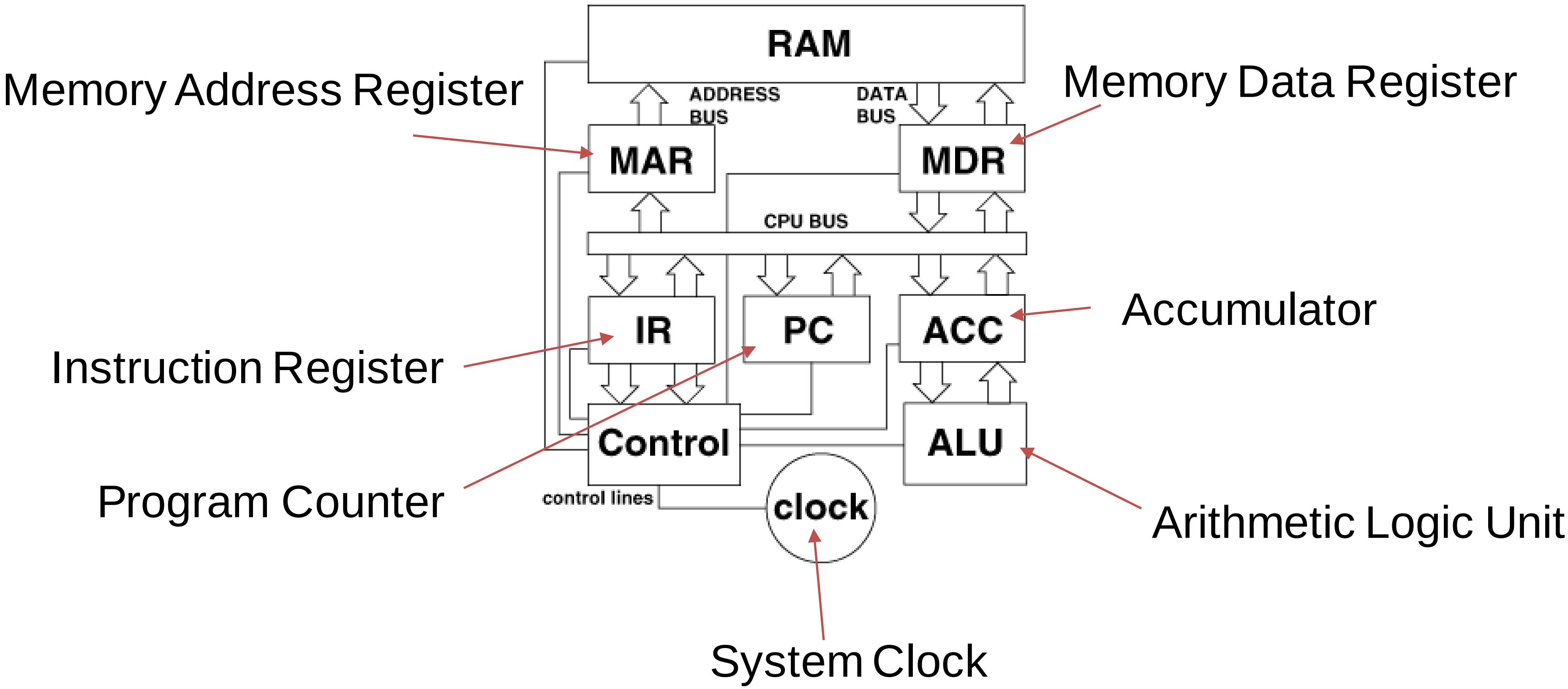




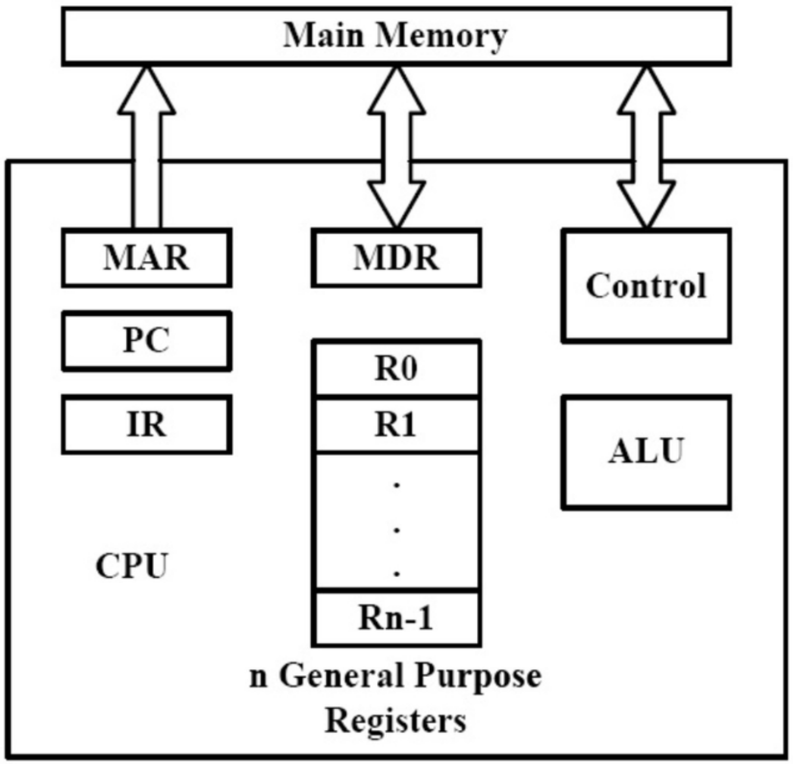


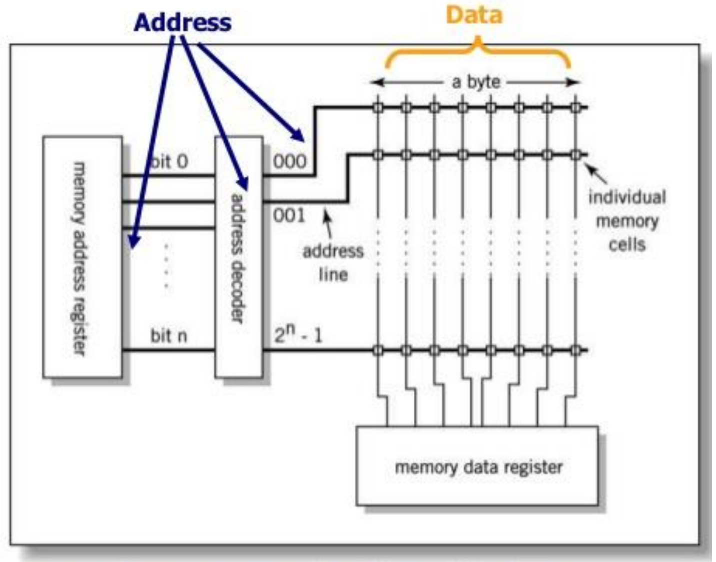
- One memory location is a collection of 8 bits
- Each unique address maps to a Byte

CPU and Special Registers



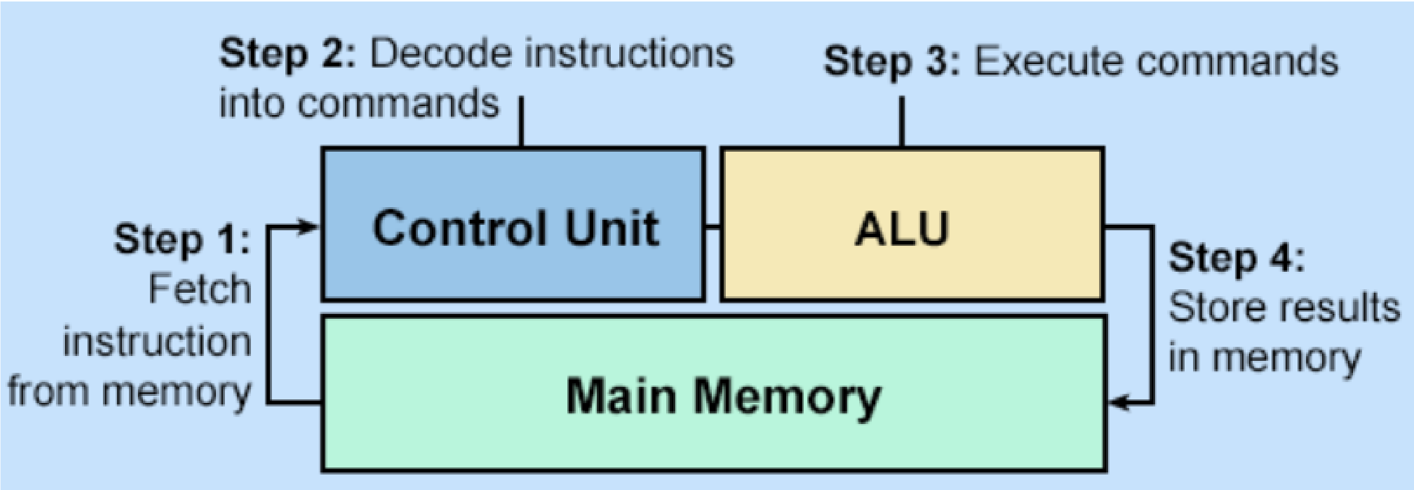
Processor Memory Interaction

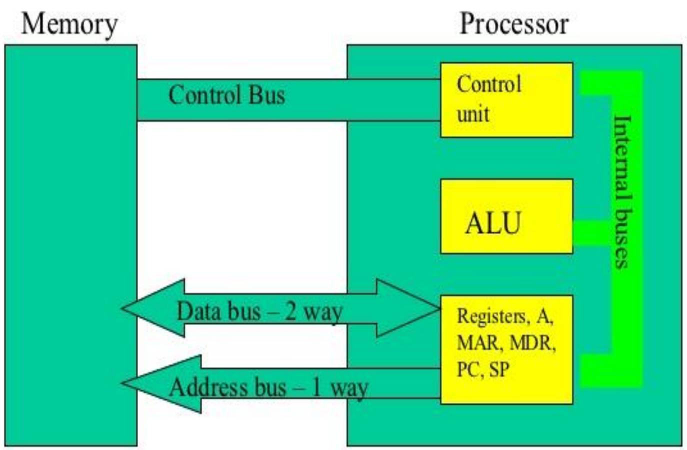


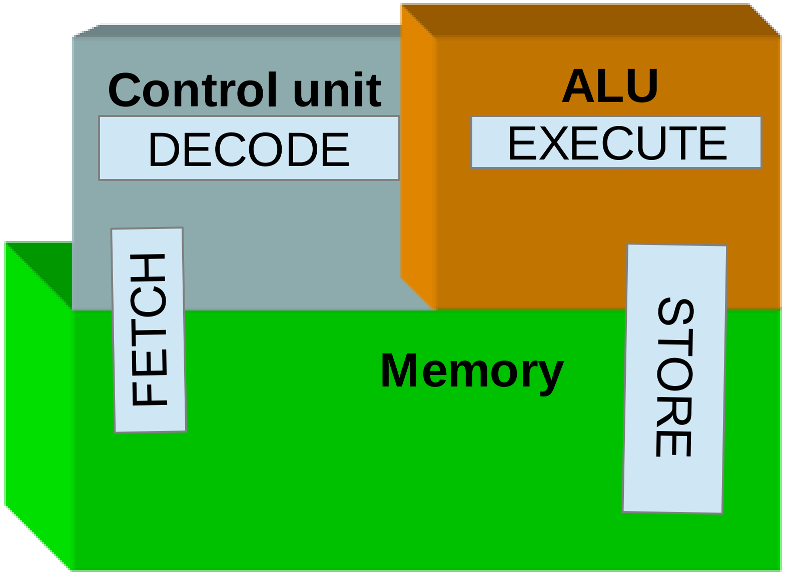


- Address generated by the CPU comes out to address bus through MAR.
- Address bus is connected to Address decoder.

Instruction Execution

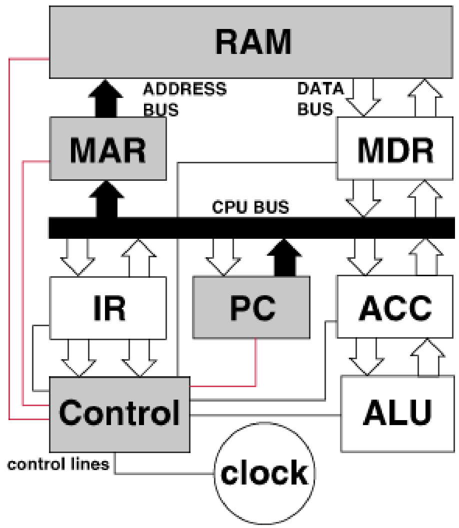




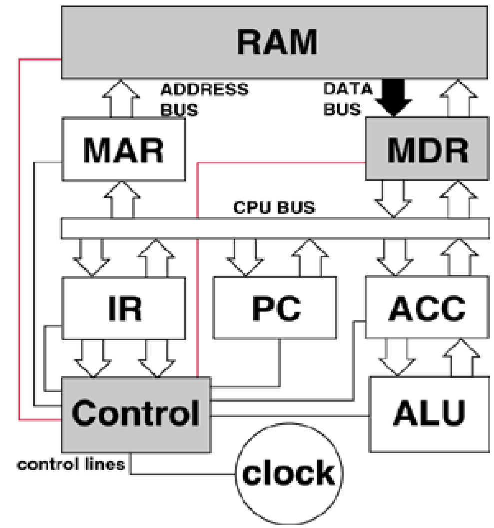


Instruction Execution Sequence

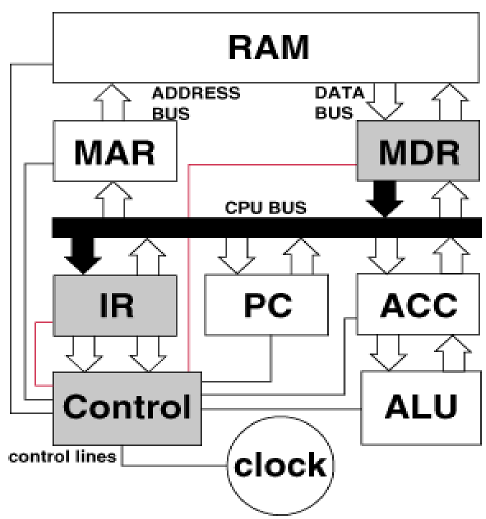
STEP-1
Issue Address



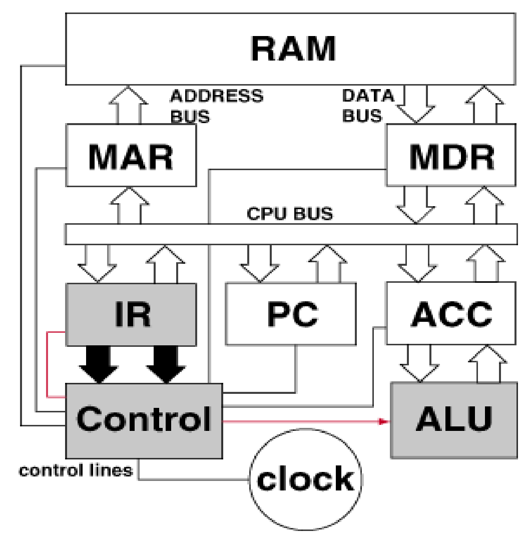
STEP-2
Fetch Instruction



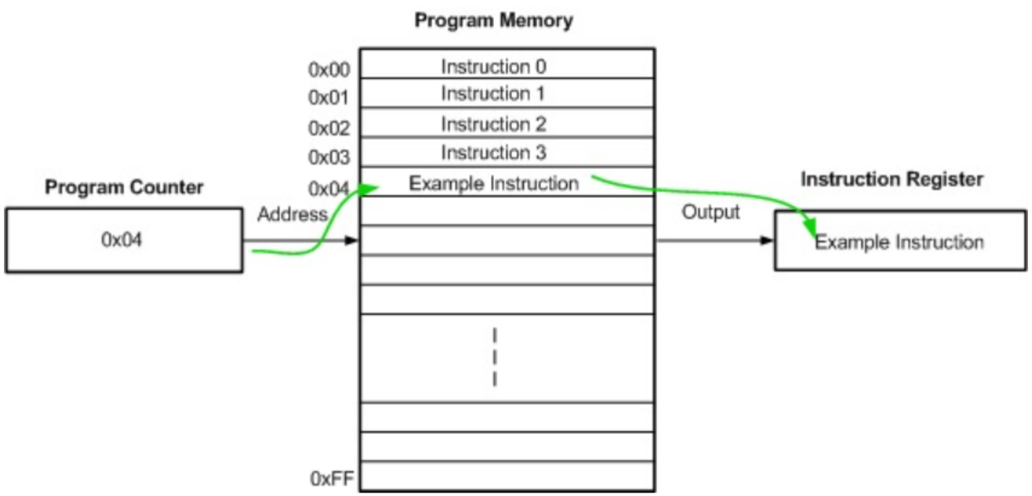
STEP-3
Decode Instruction

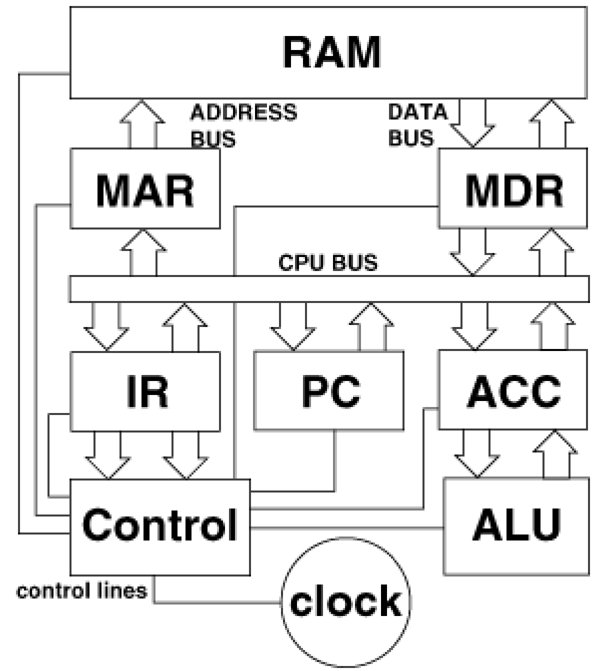


STEP-4
Execute



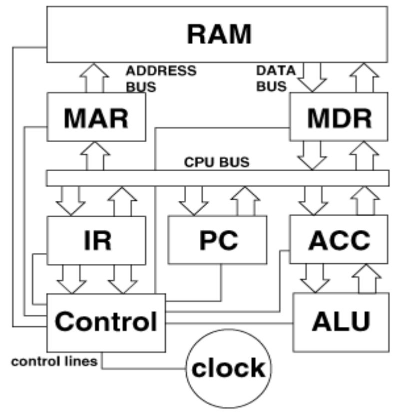
Role of PC and IR

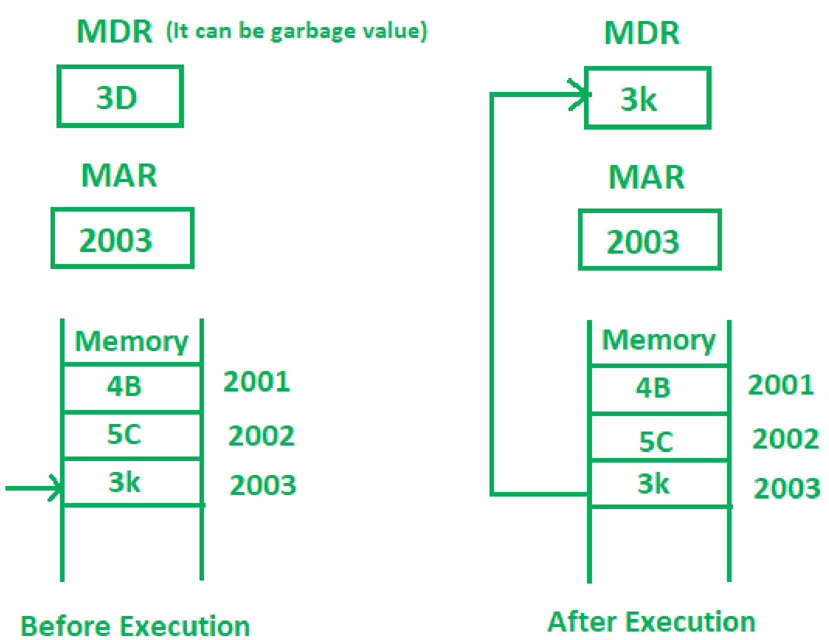




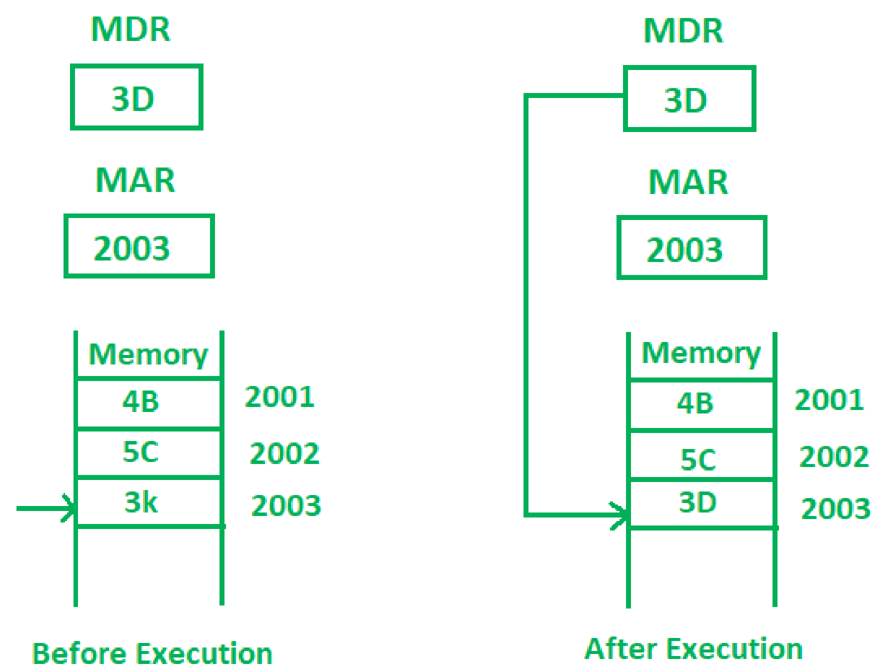
Program Counter	Instruction Register
Holds the memory address of the next instruction to be executed.	Holds the actual instruction.
After fetching a instruction the address value is increased by 1.	After fetching a instruction, it contains the next instruction.
A CPU locates the instructions in the memory from here.	A CPU decodes and executes the instructions from here.

Role of MAR and MDR





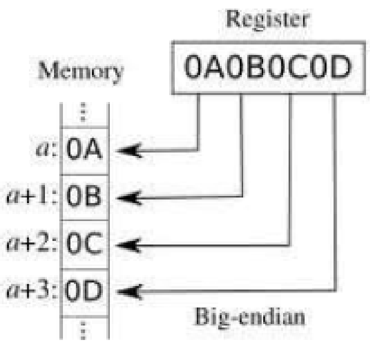
Memory Read Operation

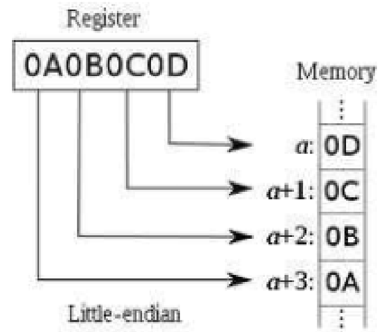


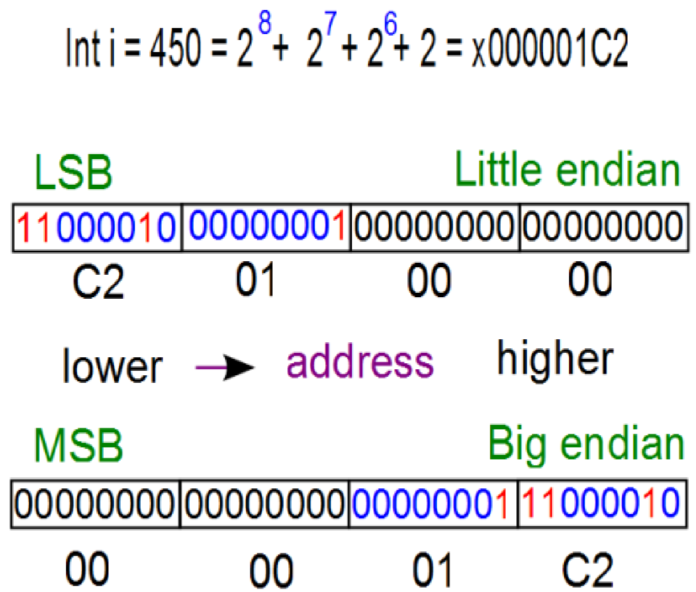
Memory Write Operation

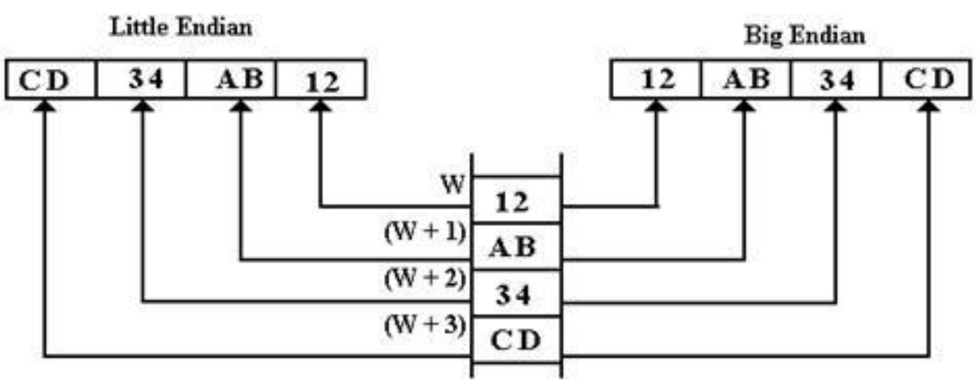
Byte Ordering

Big Endian vs. Little Endian









Byte Ordering

A 32-bit number 0x23456789 has to be written to memory (collection of bytes) from location 2000 onwards. What is the value stored in location 2001, if the system follows, (a) big endian format and (b) little endian format?

Big Endian

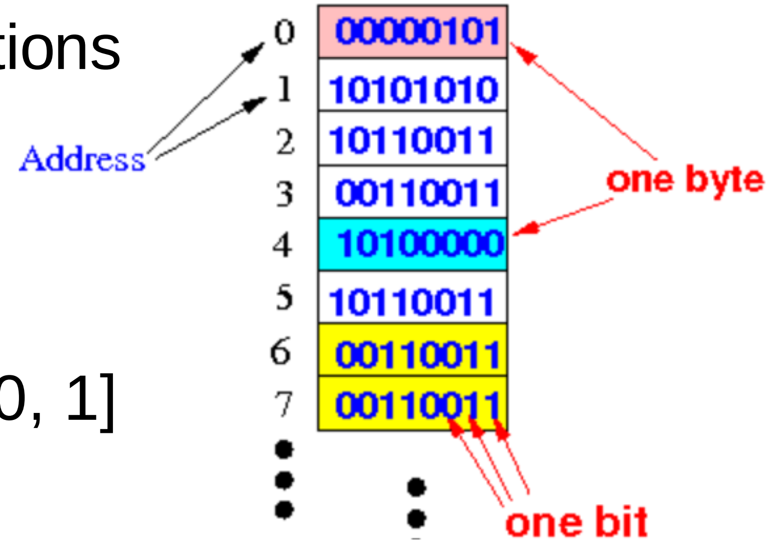
Address	**Contents**
2000	0x23
2001	0x45
2002	0x67
2003	0x89
2004	

Little Endian

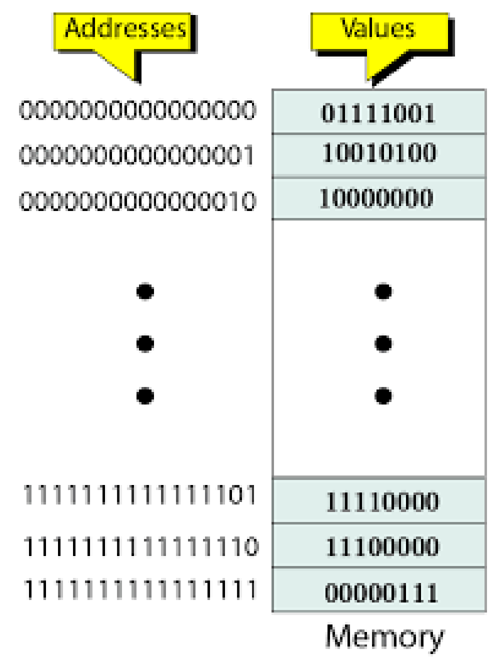
Address	**Contents**
2000	0x89
2001	0x67
2002	0x45
2003	0x23
2004	

Address Bits and Addressability

- Memory consist of collection of locations
- 1 location is generally 1 Byte
- Each location has a unique address
- 1 bit can uniquely address 2 Bytes [0, 1]
- 2 bits can uniquely address 4 Bytes
  00, 01, 10, 11 are the 4 unique addresses



- As number of bits in address increases physical address space also increases.



Larger Memory Naming Conventions

8 bits = 1 Byte

1024 Bytes = 1 Kilobyte (KB)

1024 Kilobyte = 1 Megabyte (MB)

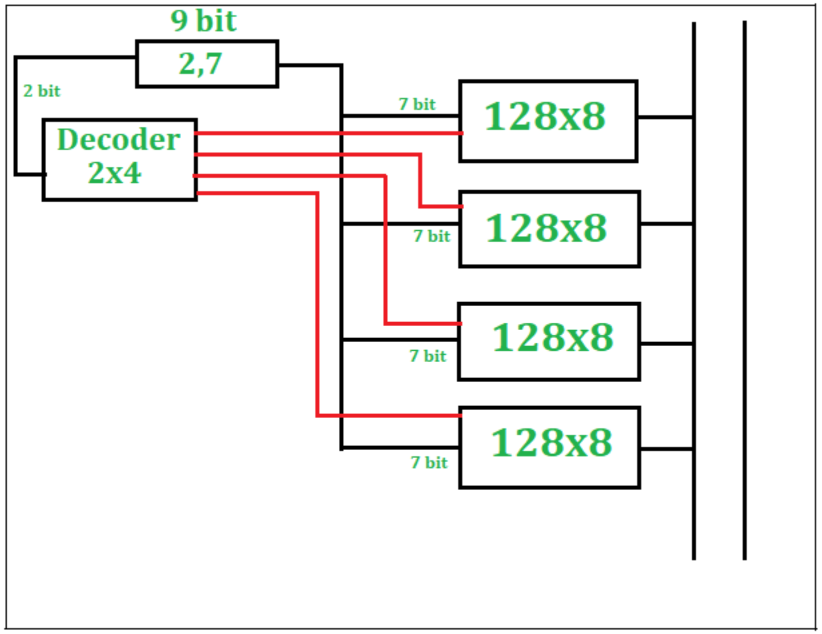
1024 Megabyte = 1 Gigabyte (GB)

1024 Gigabyte = 1 Terabyte (TB)

- Kilobyte, megabyte, gigabyte
- How many bytes can be addressed using 8-bit address?
- 2^8=256 Bytes
- If a memory is 64KB, how many bits are needed to address it?
- 64KB→$2^6.2^{10}$ Bytes → 2^{16} Bytes → 16 bit address
- If a memory is 4GB, how many bits are needed to address it?
- 4GB→$2^2.2^{30}$ Bytes → 2^{32} Bytes → 32 bit address

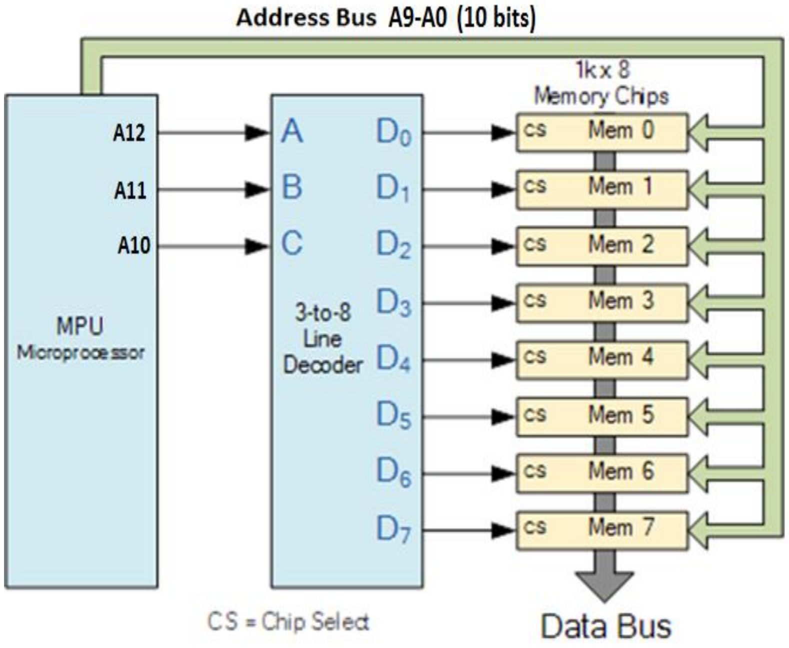
How to make large memory?

- Given a 128 B memory units, how will you create 512 B memory?
- 512 B → 9 bits for address but memory units are 128 B, which needs 7 bit address.



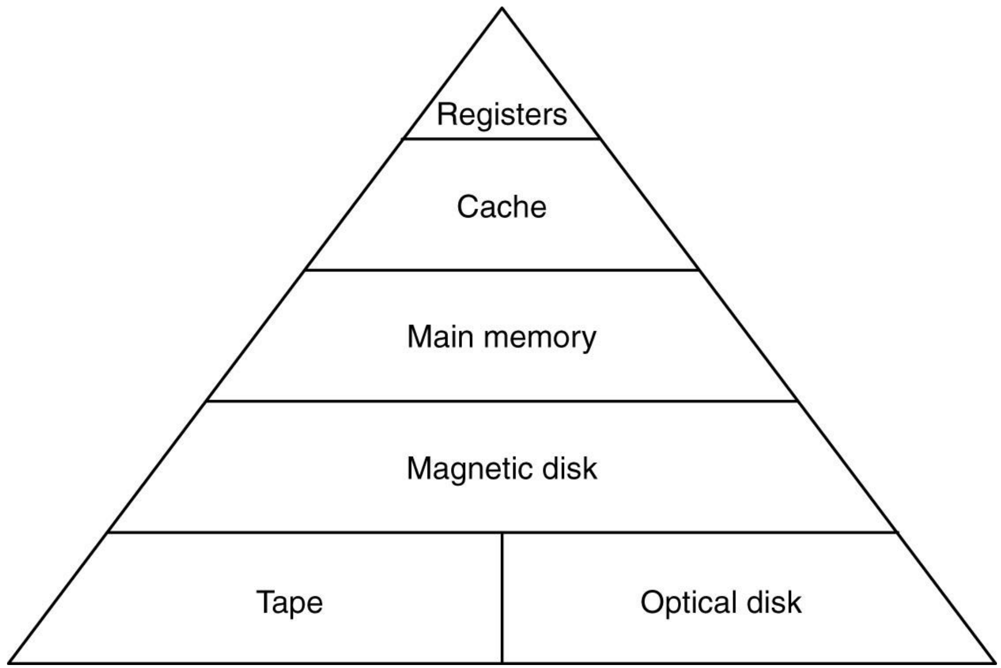
Design of 512 x 8 RAM

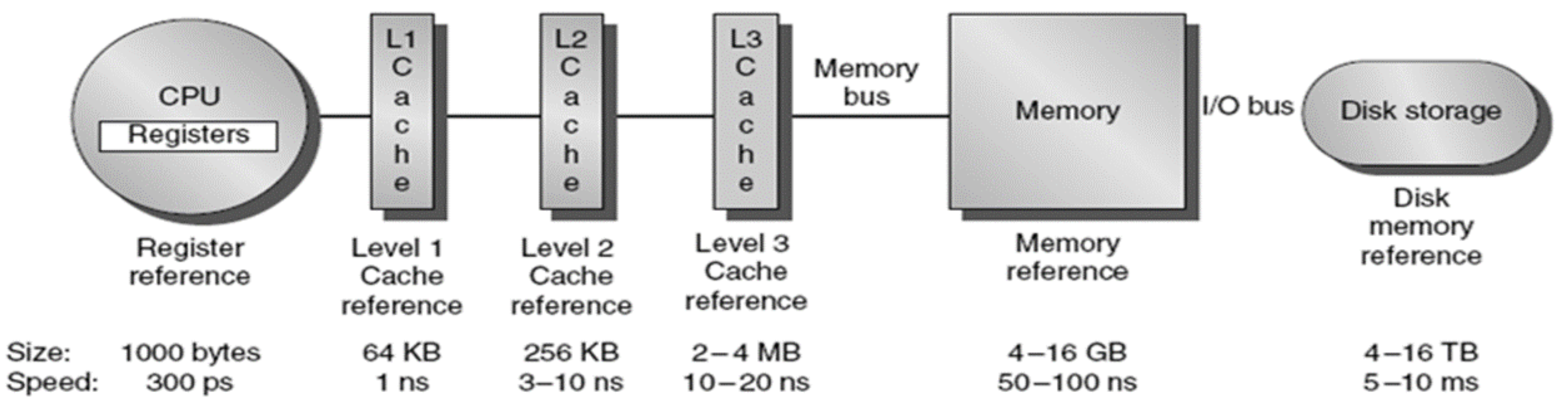
512 B = 4 x 128 B chips



8 KB = 8 x 1 KB chips

Memory Hierarchies





Thanks